Homework Assignment 1

Math 311, Fall 2026

Rutgers University

Exercise 1.2.5

(a) Suppose $x \in (A \cap B)^c$. Then, $x \notin A \cap B$. So either $x \notin A$ or $x \notin B$. In other words, $x \in A^c$ or $x \in B^c$. Then, $x \in A^c \cup B^c$. Therefore, $(A \cap B)^c \subseteq A^c \cup B^c$.

(b) Suppose $x \in A^c \cup B^c$. So $x \notin A$ or $x \notin B$. Furthermore, x couldn't be in both sets at the same time. So $x \notin A \cap B$, or $x \in (A \cap B)^c$. Therefore, $A^c \cup B^c \subseteq (A \cap B)^c$. Combining this with our result from part (a), we have that $A^c \cup B^c = (A \cap B)^c$.

(c) Suppose $x \in (A \cup B)^c$. So $x \notin A \cup B$. Then, $x \notin A$ and $x \notin B$. In other words, $x \in A^c$ and $x \in B^c$. So $x \in A^c \cap B^c$. Thus, $(A \cup B)^c \subseteq A^c \cap B^c$.

Next, suppose $y \in A^c \cap B^c$. Then, $y \notin A$ and $y \notin B$. That is, $y \notin A \cup B$. So $y \in (A \cup B)^c$. Then, $A^c \cap B^c \subseteq (A \cup B)^c$.

Therefore, $A^c \cap B^c = (A \cup B)^c$.

Exercise 1.2.9

(a) $f^{-1}(A) = [-2, 2]$ and $f^{-1}(B) = [-1, 1]$.

Notice that $A \cap B = [0, 1]$, and $f^{-1}(A \cap B) = [-1, 1]$. Similarly, $f^{-1}(A) \cap f^{-1}(B) = [-1, 1]$ as well. Thus, $f^{-1}(A \cap B) = f^{-1}(A) \cap f^{-1}(B)$

Consider $A \cup B = [-1, 4]$. Then,

$$\begin{aligned} f^{-1}(A \cup B) &= f^{-1}([-1, 4]) \\ &= f^{-1}([0, 4]) \\ &= [-2, 2]. \end{aligned}$$

Similarly, $f^{-1}(A) \cup f^{-1}(B) = [-2, 2]$. Thus, $f^{-1}(A \cup B) = f^{-1}(A) \cup f^{-1}(B)$.

(b) Let $x \in \mathbb{R}$. Then,

$$\begin{aligned} x \in g^{-1}(A \cap B) &\iff g(x) \in A \cap B \\ &\iff g(x) \in A \text{ and } g(x) \in B \\ &\iff x \in g^{-1}(A) \text{ and } x \in g^{-1}(B) \\ &\iff x \in g^{-1}(A) \cap g^{-1}(B). \end{aligned}$$

Therefore,

$$g^{-1}(A \cap B) = g^{-1}(A) \cap g^{-1}(B).$$

Similarly,

$$\begin{aligned} x \in g^{-1}(A \cup B) &\iff g(x) \in A \cup B \\ &\iff g(x) \in A \text{ or } g(x) \in B \\ &\iff x \in g^{-1}(A) \text{ or } x \in g^{-1}(B) \\ &\iff x \in g^{-1}(A) \cup g^{-1}(B). \end{aligned}$$

Therefore,

$$g^{-1}(A \cup B) = g^{-1}(A) \cup g^{-1}(B).$$

Exercise 1.2.10

(a) False. Consider $a = b$. Then, for every $\epsilon > 0$,

$$a < b + \epsilon.$$

But, since $a = b$, the statement $a < b$ must be false. Thus, the if the and only if does not hold both ways.

(b) False, same as the previous counter example.

(c) True. Suppose $a \leq b$ Since $\epsilon > 0$. It follows that

$$b < b + \epsilon.$$

So,

$$a \leq b < b + \epsilon.$$

That is,

$$a \leq b \implies a < b + \epsilon \mid \forall \epsilon > 0.$$

Now, suppose that $a < b + \epsilon$ for every $\epsilon > 0$. For sake of contradiction, suppose $a > b$. Then $a - b > 0$. Choose

$$\epsilon = \frac{a-b}{2}.$$

Then

$$a < b + \frac{a-b}{2} = \frac{a+b}{2}$$

But, since $a > b$,

$$a > \frac{a+b}{2}.$$

Contradicting our previous inequality, so $a \leq b$. Therefore,

$$a < b + \epsilon \mid \forall \epsilon > 0 \implies a \leq b.$$

.

Hence,

$$a \leq b \iff a < b + \epsilon \mid \forall \epsilon > 0.$$

Exercise 1.2.13

(a) Let S_n denote the statement that

$$(A_1 \cup A_2 \cup \cdots \cup A_n)^c = A_1^c \cap A_2^c \cap \cdots \cap A_n^c.$$

(Base Case) Then S_1 says $(A_1)^c = A_1^c$, which is obviously true.

(Inductive Step) We suppose S_k for an arbitrary $k \in \mathbb{N}$. That is,

$$(A_1 \cup A_2 \cup \cdots \cup A_k)^c = A_1^c \cap A_2^c \cap \cdots \cap A_k^c.$$

Consider

$$(A_1 \cup A_2 \cup \cdots \cup A_k \cup A_{k+1})^c$$

Then group the first k sets,

$$((A_1 \cup A_2 \cup \cdots \cup A_k) \cup A_{k+1})^c.$$

Apply 2-element De Morgan's Law from Exercise 1.2.5

$$(A_1 \cup A_2 \cup \cdots \cup A_k)^c \cap A_{k+1}^c.$$

Substitute S_k

$$A_1^c \cap A_2^c \cap \cdots \cap A_k^c \cap A_{k+1}^c.$$

That is,

$$(A_1 \cup A_2 \cup \cdots \cup A_k \cup A_{k+1})^c = A_1^c \cap A_2^c \cap \cdots \cap A_k^c \cap A_{k+1}^c.$$

So S_{k+1} holds. Therefore, by mathematical induction, S_n is true for every $n \in \mathbb{N}$.

(b) Let $B_i = (0, \frac{1}{i})$. Then for any $n \in \mathbb{N}$,

$$\bigcap_{i=1}^{n} B_i = (0, \frac{1}{n}).$$

Since $n \in \mathbb{N}$, we can always find a larger natural number $n+1$. So $\frac{1}{n+1}$ will be inside the interval $(0, \frac{1}{n})$. Thus, the intersection is always nonempty.

Now suppose

$$x \in \bigcap_{i=1}^{\infty} B_i.$$

Then $0 < x < \frac{1}{i}$ for every $i \in \mathbb{N}$. Since $x > 0$, by the Archimedean Principle, there exists $i \in \mathbb{N}$ such that $\dfrac{1}{i} < x$. This contradicts $x < \dfrac{1}{i}$. Hence, no such x exists. That is,

$$\bigcap_{i=1}^{\infty} B_i = \emptyset.$$

(c) Suppose

$$x \in \left(\bigcup_{i=1}^{\infty} A_i\right)^c$$

Equivalently,

$$x \notin \bigcup_{i=1}^{\infty} A_i$$

That is, $x \notin A_i$ for every $i \in \mathbb{N}$. Then $x \in A_i^c$ for every $i \in \mathbb{N}$. Therefore,

$$x \in \bigcap_{i=1}^{\infty} A_i^c$$

So

$$\left(\bigcup_{i=1}^{\infty} A_i\right)^c \subseteq \bigcap_{i=1}^{\infty} A_i^c$$

Now, suppose

$$y \in \bigcap_{i=1}^{\infty} A_i^c$$

Then $y \in A_i^c$ for every $i \in \mathbb{N}$. Likewise, $y \notin A_i$ for every $i \in \mathbb{N}$. That is,

$$y \notin \bigcup_{i=1}^{\infty} A_i$$

So,

$$y \in \left(\bigcup_{i=1}^{\infty} A_i\right)^c$$

Thus,

$$\bigcap_{i=1}^{\infty} A_i^c \subseteq \left(\bigcup_{i=1}^{\infty} A_i\right)^c$$

Combining the two inclusions, we get

$$\left(\bigcup_{i=1}^{\infty} A_i\right)^c = \bigcap_{i=1}^{\infty} A_i^c.$$

Exercise 1.3.9

(a) Assume $\sup A < \sup B$. For the sake of contradiction, assume $\sup A$ is an upper bound for B. Then,

$$\sup B \leq \sup A,$$

contradicting $\sup A < \sup B$. Therefore, there exists $b \in B$ such that

$$b > \sup A.$$

Furthermore,

$$a \leq \sup A < b \text{ for every } a \in A.$$

Thus, b is an upper bound for A.

(b) Consider

$$A = B = (0, 1).$$

Then

$$\sup A = \sup B = 1,$$

and we have

$$\sup A \leq \sup B.$$

But, no $b \in B$ can be an upper bound for A. For every $b \in B$, we can simply choose

$$a = \frac{b+1}{2}.$$

Then $b < a < 1$, so $a \in A$ but $b < a$. Therefore, b cannot be an upper bound for A.

Exercise 1.3.10

(a) Assume A and B are non-empty, disjoint sets with

$$A \cup B = \mathbb{R}$$

and

$$a < b \text{ for every } a \in A, b \in B.$$

Choose an arbitrary element $b_0 \in B$, then $b_0 > a$ for every $a \in A$. That is, b_0 is an upper bound for A. Since A is non-empty and bounded above, the Axiom of Completeness guarantees us a least upper bound, call it

$$c = \sup A$$

Since c is an upper bound for A,

$$a \leq c \text{ for every } a \in A.$$

Furthermore, because c is the least upper bound on A and every $b \in B$ is an upper bound for A,

$$c \leq b \text{ for every } b \in B.$$

Therefore,

$$a \leq c \leq b$$

for every $a \in A$ and $b \in B$.

(b) Assume the Cut Property holds for $\mathbb{R}$. Let $E \subseteq \mathbb{R}$ be a nonempty set that is bounded above. Define

$$A = \{x \in \mathbb{R} : x \text{ is not an upper bound for } E\}$$

and

$$B = \{x \in \mathbb{R} : x \text{ is an upper bound for } E\}.$$

Then $A \cup B = \mathbb{R}$, $A \cap B = \emptyset$, and $B \neq \emptyset$ since E is bounded above. Choose some $e \in E$. Then $e - 1 < e$. So $e - 1$ is not an upper bound on E. So $e - 1 \in A$ and $A \neq \emptyset$. Choose some $a \in A$ and $b \in B$. Since a is not an upper bound on E, there exists some $x \in E$ such that

$$a < x.$$

Since b is an upper bound on E,

$$x \leq b.$$

Thus,

$$a < x \leq b,$$

so

$$a < b.$$

Therefore the Cut Property holds. So there exists some $c \in \mathbb{R}$ such that

$$a \leq c \leq b$$

for every $a \in A$ and $b \in B$.

We want to prove that $c = \sup E$.

Suppose, for sake of contradiction, that c is not an upper bound on E. Then there exists some $x \in E$ such that

$$x > c.$$

Choose

$$y = \frac{c + x}{2}.$$

Then

$$c < y < x.$$

Since $x \in E$ and $y < x$, y cannot be upper bound on E, so $y \in A$. But the Cut Property says that every $a \in A$ satisfies $a \leq c$, so

$$y \leq c,$$

contradicting $c < y$. Thus, c must be an upper bound for E, so $c \in B$. By Cut Property,

$$c \leq b$$

for every $b \in B$. That is, c is the least upper bound of E, or

$$c = \sup E.$$

Therefore, Cut Property implies the Axiom of Completeness.

(c) Define

$$A = \{q \in \mathbb{Q} : q < \sqrt{2}\}$$

and

$$B = \{q \in \mathbb{Q} : q > \sqrt{2}\}$$

Then $A \cup B = \mathbb{Q}$, $A \cap B = \emptyset$, and $a < b$ for every $a \in A$ and $b \in B$. If the Cut Property were true in $\mathbb{Q}$, there would be some $c \in \mathbb{Q}$ such that

$$a \leq c \leq b$$

for every $a \in A$ and $b \in B$. But the only number that can sit between all rationals below $\sqrt{2}$ and all rationals above $\sqrt{2}$ is

$$c = \sqrt{2},$$

and $\sqrt{2} \notin \mathbb{Q}$. Therefore no such rational c exists, so the Cut Property fails in $\mathbb{Q}$.